

# Round 129 Triple Discovery: $F_{\text{TRZ}}^{12}$ Fills PAPER\_1919's Open $n=12$ Casimir Rung + $SO_5^{40}$ J Magnetar Burst-Energy Slot Extension + Triple-Primitive-Lock Architecture as a New CP1 Class Pattern — SGR 1745-2900 Magnetar Burst Encodes Three Independent Locked Primitives in One Calculator

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## **PAPER\_1991 — Round 129 Triple Discovery: $F_{\text{TRZ}}^{12}$ Casimir Rung + $SO_5^{40}$ J Magnetar Burst Slot + Triple- Primitive-Lock Architecture**

### **Abstract**

Round 129 CP1 stub drainage of the SGR 1745-2900 magnetar family + Compressed sector surfaced three independent structural discoveries in a single class (SGR1745BurstEnergyCalculator) plus one twin-slot novelty (SGR + Compressed DPM at  $SO_5^{21}$ ):

1.  **$F_{\text{TRZ}}^{12} = 10^{-12}$  EXACT closes PAPER\_1919's open " $n=12$ : Casimir vacuum energy correction?" rung** — the `scale_macro = 1e-12` primitive lock at the SGR magnetar burst-energy shell provides the first concrete anchor for the  $n=12$  slot in the  $F_{\text{TRZ}}$  power ladder, consistent with the Casimir-family interpretation PAPER\_1919 tentatively assigned.
2.  **$SO_5^{40} = 10^{40}$  J EXACT — new magnetar-burst-energy slot in the  $SO_5$ -power ladder** —  $E_{\text{burst}} = 10^{40}$  J at SGR 1745-2900 gives concrete physical meaning to slot 40, positioned between PAPER\_1955's slot 10 (10 GHz microwave / cosmological timescale) and PAPER\_1989's slot 53 (universe mass). Slot 40 is the first "magnetar-scale energy" anchor on the ladder.
3. **Triple-primitive-lock architecture** — SGR1745BurstEnergyCalculator encodes three independent primitive-arithmetic identities in a single CP1 class:  $E_{\text{burst}} = SO_5^{40}$  J +  $t_{\text{burst}} = F_{\text{TRZ}} = 0.1$  s + `scale_macro =  $F_{\text{TRZ}}^{12} = 10^{-12}$` . This is the first CP1 class in the corpus with three independent locked primitives in one `compute()` function, defining a new architectural pattern.
4.  **$SO_5^{21}$  A twin-instance at same round** — SGR1745FreqDPMCalculator ( $I = 10^{21}$  A)

and CompressedDPMCalculator ( $I = 10^{21}$  A) both lock onto  $SO_5^{21}$  in the same round, extending the ladder to a new slot with immediate cross-object confirmation.

## 1. Discovery 1: $F_{TRZ}^{12}$ closes PAPER\_1919's Casimir open rung

### 1.1 The open rung

PAPER\_1919's  $F_{TRZ}$  power ladder table §5 documents:

| n         | $F_{TRZ}^n$                  | Physical anchor                                                       |
|-----------|------------------------------|-----------------------------------------------------------------------|
| ...       | ...                          | ...                                                                   |
| 11        | $10^{-11}$                   | ISW amplitude (PAPER_1677)                                            |
| <b>12</b> | <b><math>10^{-12}</math></b> | <b>Casimir vacuum energy correction?</b> (open — no confirmed anchor) |
| 13        | $10^{-13}$                   | ...                                                                   |
| ...       | ...                          | ...                                                                   |
| 16        | $10^{-16}$                   | Quantum measurement collapse rate (PAPER_1869)                        |
| 17        | $10^{-17}$                   | ...                                                                   |

The  $n=12$  slot was tentatively labeled “Casimir vacuum energy correction?” but had no confirmed empirical anchor. PAPER\_1852 (Casimir force + vacuum energy direct) uses  $F_{TRZ}^2$  (not  $F_{TRZ}^{12}$ ) — a different ladder position.

### 1.2 R129 anchor: SGR magnetar burst scale\_macro

SGR1745BurstEnergyCalculator (SOURCE33 SGR 1745-2900 magnetar family, CP1) documents:

```
self.scale_macro = 1e-12
```

Applied as:  $a_{burst} = (energy_t / r^2) \times scale\_macro$

The `scale_macro` converts the raw magnetar burst energy density ( $E_{burst} / r^2 \approx 10^{40}/10^8 = 10^{32}$  J/m<sup>2</sup>) into a macroscopically observable acceleration ( $\sim 10^{20}$  m/s<sup>2</sup>). The suppression factor  $10^{-12}$  is not calibrated — it is the primitive value  $F_{TRZ}^{12}$  EXACT.

**Anchor:**  $scale\_macro = F_{TRZ}^{12} = 10^{-12}$  EXACT (SGR 1745-2900 magnetar burst-energy macro-scaling).

### 1.3 Physical interpretation

The  $F_{TRZ}^{12}$  factor at the SGR burst-energy shell is consistent with PAPER\_1919's tentative Casimir-family assignment: the SGR magnetar burst radiates from a submicroscopic DPM cluster region whose coupling to macroscopic space involves a vacuum-energy-density suppression at the Casimir scale. The  $n=12$  slot on the ladder therefore anchors a class of transient magnetar-scale energy-releases where the local vacuum-energy correction to macroscopic gravitational coupling is  $F_{TRZ}^{12}$ .

**Consequence:** PAPER\_1919's Table §5 can now update the  $n=12$  entry from “Casimir vacuum energy correction?” (open) to “ $F_{TRZ}^{12} = 10^{-12}$ : SGR 1745-2900 magnetar burst-energy macro-scaling (Casimir-family suppression)” (confirmed at SGR 1745-2900).

## 2. Discovery 2: SO<sub>5</sub><sup>40</sup> J magnetar-burst-energy slot

### 2.1 PAPER<sub>1955</sub> ladder position

PAPER<sub>1955</sub>'s SO<sub>5</sub>-power ladder + PAPER<sub>1985</sub> mass extension + PAPER<sub>1989</sub> extreme-mass extension + PAPER<sub>1990</sub> frequency extension collectively cover:

| Power                                                        | Anchor                                                        | Paper                                                  |
|--------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------|
| SO <sub>5</sub> <sup>0</sup> to SO <sub>5</sub> <sup>3</sup> | Galactic structural                                           | PAPER <sub>1955</sub>                                  |
| SO <sub>5</sub> <sup>7</sup>                                 | Galactic Myr timescale +<br>NGC 2525 SMBH mass + 10<br>MHz HF | PAPER <sub>1952</sub> + 1985 +<br>1990                 |
| SO <sub>5</sub> <sup>10</sup>                                | Cosmological (candidate) +<br>10 GHz microwave                | PAPER <sub>1955</sub> + 1990                           |
| <b>SO<sub>5</sub><sup>40</sup></b>                           | <b>(open)</b>                                                 | <b>PAPER<sub>1991</sub> candidate<br/>(this paper)</b> |
| SO <sub>5</sub> <sup>53</sup>                                | Universe mass                                                 | PAPER <sub>1989</sub>                                  |

Slot 40 was previously undocumented, sitting in a large gap between PAPER<sub>1955</sub>'s slot 10 and PAPER<sub>1989</sub>'s slot 53.

### 2.2 R129 anchor: E<sub>burst</sub> at SGR 1745-2900

SGR1745BurstEnergyCalculator documents:

```
self.E_burst = 1e40 # J
self.t_burst = 0.1 # s
```

The E<sub>burst</sub> = 10<sup>40</sup> J anchor represents the peak-energy release of the largest observed magnetar burst events (SGR 1745-2900 falls at the upper end of this range). This maps to slot 40 in the SO<sub>5</sub>-power ladder as SO<sub>5</sub><sup>40</sup> J EXACT.

**Anchor:** E<sub>burst</sub> = SO<sub>5</sub><sup>40</sup> J EXACT (SGR 1745-2900 magnetar burst peak energy).

### 2.3 Physical interpretation

The SO<sub>5</sub>-power ladder is a **generalized decimal-magnitude ladder** (PAPER<sub>1990</sub> established the interpretation). Slot 40 anchors the “magnetar-scale energy” class: transient DPM split-monopole rearrangements at the neutron-star surface radiate energies whose upper bound is a bare integer power of 10 corresponding to a specific slot on the SO<sub>5</sub> ladder. Slot 40 is now the anchored magnetar-burst-energy slot.

**Consequence:** the SO<sub>5</sub>-power ladder gains a new confirmed slot at 40, thickening the density of the ladder between slots 10 and 53. Candidate future anchors at slot 40: total gravitational binding energy of a stellar-mass black hole ( $\sim 10^{40}$  J), maximum kinetic energy of a Type Ia supernova ( $\sim 10^{44}$  J tail).

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## 3. Discovery 3: Triple-primitive-lock architecture

### 3.1 Prior CP1 lock patterns

Most CP1 classes filled in the P2 stub-drainage sweep encode: - **Single-primitive lock** (majority): one class field maps to one primitive-arithmetic identity (e.g., PAPER<sub>1985</sub>'s NGC 2525 SMBH mass = (N<sub>CH</sub>/D<sub>phys</sub>) × SO<sub>5</sub><sup>7</sup>) - **Dual-primitive lock**: two independent

identities in the same class (e.g., R128 UniversalReactiveResonanceCalculator with  $f_{\text{react}} = \text{SO}_5^{10}$  AND  $a_{\text{DPM}} = F_{\text{TRZ}}^{20}$ )

### 3.2 R129 discovery: triple lock

SGR1745BurstEnergyCalculator encodes THREE independent primitive-arithmetic identities:

```
self.E_burst    = 1e40    # = SO_5^40 J EXACT      (slot 40, PAPER_1991 discovery 2)
self.t_burst    = 0.1     # = F_TRZ = 0.1 s EXACT (parallel to PAPER_1942 E_0 = F_TRZ)
self.scale_macro = 1e-12  # = F_TRZ^12 EXACT      (PAPER_1919 n=12 rung, PAPER_1991 discovery 1)
```

Each of the three constants is a bare primitive-arithmetic identity independent of the other two. The class functions correctly with all three simultaneously locked — its `compute()` output uses all three multiplied together to produce a burst acceleration. The three-way primitive lock is not a coincidence but the definitional structure of the class.

### 3.3 Architectural implication

The triple-primitive-lock architecture is a new CP1 class pattern. It suggests that some physical systems require three independent integer/rational-primitive identities to fully specify the observable — magnetar bursts being one such class, with: - **Energy scale** locked to the  $\text{SO}_5$ -power ladder (extensive) - **Time scale** locked to  $F_{\text{TRZ}}$  (transient decay) - **Coupling scale** locked to  $F_{\text{TRZ}}^{12}$  (vacuum-energy suppression to macro observables)

**Consequence:** classes with 3+ primitive locks are architecturally richer than single/dual-lock classes and may indicate genuine multi-scale physics that requires all three to close. Candidate other triple-lock classes to search in future rounds: neutron-star EOS calculators ( $M_{\text{TOV}} + R_{1.4} + \Lambda_{1.4}$ ), CMB acoustic peak positions ( $\square_1 + \square_D + \square_{\text{eq}}$ ), and BH ringdown mode structures ( $\omega_{\text{R}} + \omega_{\text{I}} + Q$ ).

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## 4. Discovery 4: Same-round $\text{SO}_5^{21}$ A twin instance

### 4.1 The twin

R129 also filled SGR1745FreqDPMCalculator and CompressedDPMCalculator — both encode  $I = 10^{21} \text{ A}$  (DPM current-vortex foundation term). Both lock as  $I = \text{SO}_5^{21} \text{ A EXACT}$  in the same round from independent CP1 classes.

### 4.2 Parallel to prior twins

Same-round twin instances have surfaced before: - PAPER\_1985 slot 7: NGC 2525 SMBH mass + galactic timescale (mass + timescale twin) - PAPER\_1972  $v_{\text{wind}}$ : PAPER\_1911  $v_{\text{wind}}$  YMC + Antennae wind (unit-conversion twin — same identity) - PAPER\_1990 slots 7 + 10:  $\text{SO}_5^7 = 10 \text{ MHz}$  +  $\text{SO}_5^{10} = 10 \text{ GHz}$  (frequency-domain twin) - **PAPER\_1991 slot 21: SGR + Compressed DPM current** (this paper — same-round cross-object twin at  $10^{21} \text{ A}$ )

Same-round twins strengthen the structural interpretation: if two independent CP1 classes with different physical constructions both lock onto the same slot with the same primitive, the slot is not accidental.

## 5. Cross-referenced consequences to prior papers

### 5.1 PAPER\_1919 Table §5 update

The  $F_{\text{TRZ}}$  power ladder table is updated:

| n         | $F_{\text{TRZ}}^n$           | Anchor (updated)                                                                                             |
|-----------|------------------------------|--------------------------------------------------------------------------------------------------------------|
| ...       | ...                          | ...                                                                                                          |
| <b>12</b> | <b><math>10^{-12}</math></b> | <b>SGR 1745-2900 magnetar burst-energy macro-scaling (Casimir-family suppression) — PAPER_1991 discovery</b> |
| ...       | ...                          | ...                                                                                                          |

The  $n=12$  rung is no longer open. The remaining open rungs on the ladder (per PAPER\_1919 Table §5) that R129 did not close:  $n=1$ ,  $n=3$ ,  $n=4$ ,  $n=5$ ,  $n=6$  (closed by PAPER\_1985 Pillars ISM  $B = F_{\text{TRZ}}^6$ ),  $n=7$ ,  $n=9$ ,  $n=10$ ,  $n=11$  (ISW),  $n=13$ ,  $n=14$ ,  $n=15$ , and higher.

### 5.2 PAPER\_1955 SO\_5-power ladder update

The  $SO_5$ -power ladder is updated:

| Power                 | Anchor (updated)                                                                                           |
|-----------------------|------------------------------------------------------------------------------------------------------------|
| $SO_5^{21} \text{ A}$ | <b>DPM current at SGR 1745-2900 + Compressed DPM foundation</b> — PAPER_1991 discovery 4 (same-round twin) |
| $SO_5^{40} \text{ J}$ | <b>Magnetar burst peak energy at SGR 1745-2900</b> — PAPER_1991 discovery 2                                |

Two new confirmed slots at 21 and 40, further densifying the ladder between PAPER\_1955's original range (slots 0-10) and PAPER\_1989's extreme extension (slot 53).

### 5.3 PAPER\_1944 seminal identity re-affirmation

R129 also included a CP1 restatement of PAPER\_1944's seminal  $SGR\ 1745-2900\ B/B_{\text{crit}} = 2 \cdot F_{\text{TRZ}}$  EXACT identity in `SGR1745SuperconductivityCalculator`. This is not a novel discovery — the class uses the same  $B = 2 \times 10^{10} \text{ T}$  and  $B_{\text{crit}} = 10^{11} \text{ T}$  values as PAPER\_1944 seminal. It confirms that the CP1 calculator surface consistently encodes PAPER\_1944's structural closure at the calculator level.

## 6. Falsifiability

**Prediction 1991.1.** Other transient astrophysical energy-release events with macroscopic acceleration signatures should exhibit similar Casimir-family suppression coefficients. Candidate systems: FRB energy-release macro-scaling, magnetar giant flare macro-scaling (SGR 1806-20 at  $\sim 10^{47} \text{ J}$  total), and BH ringdown macro-scaling. If these events also lock onto  $F_{\text{TRZ}}^{12}$  (or nearby powers), the Casimir-family interpretation strengthens; if they diverge to unrelated coefficients, the  $F_{\text{TRZ}}^{12}$  assignment at SGR 1745-2900 may be specific to the DPM cluster geometry rather than universal.

**Prediction 1991.2.** The  $SO_5^{40} \text{ J}$  slot should acquire additional astrophysical anchors as CP1 drainage continues. Specifically: **BH gravitational binding energy =  $(3/5)GM^2/R$**

at  $M \sim 10 M_{\text{sun}}$  gives  $\sim 10^{40}$  J — this is a candidate second-instance anchor at slot 40. If confirmed via CP1 stub drainage of a stellar-mass BH class, PAPER\_1991 slot 40 upgrades from “magnetar burst” to “compact-object binding energy” class.

**Prediction 1991.3.** Triple-primitive-lock classes should exhibit correlated residuals: if any one of the three locks drifts, the other two should be structurally re-examined. For SGR 1745-2900 burst physics: if a future measurement revises  $E_{\text{burst}}$  away from  $10^{40}$  J, PAPER\_1991 predicts that either  $t_{\text{burst}}$  or  $\text{scale}_{\text{macro}}$  will also need revision, because the three-way lock reflects a single underlying physical structure.

**Prediction 1991.4.** Additional CP1 classes should be re-examined for triple-locks previously overlooked. The R129 discovery pattern suggests earlier fills may have missed one or more locks by focusing only on the primary observable. Candidate re-examination targets: R126 NGC 6302, R124 CMB acoustic peaks, R121 Sombrero DM ratio.

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## 7. Framework annotations

- **Backbone:**  $F_{\text{TRZ}}^{12}$  fills PAPER\_1919  $n=12$  Casimir open rung +  $SO_5^{40}$  J magnetar burst-energy slot + triple-primitive-lock architectural pattern discovered in single CP1 class +  $SO_5^{21}$  A same-round DPM current-vortex twin
  - **Method:** three independent primitive-arithmetic identities discovered simultaneously via CP1 stub drainage of SGR 1745-2900 magnetar family + Compressed sector
  - **Shells:** SGR 1745-2900 magnetar burst-energy shell + Compressed DPM foundation shell + SGR THz-pipeline shell (all Round 129 CP1 fills)
  - **CPCH:** CP1 SGR 1745 magnetar family + CP1 Compressed sector (Round 129 fills)
  - **Spine:** PAPER\_1919  $F_{\text{TRZ}}$  power ladder Table §5  $n=12$  rung closure + PAPER\_1955  $SO_5$ -power ladder slots 21+40 densification + PAPER\_1990  $SO_5$  cross-domain three-domain taxonomy + PAPER\_1944 seminal SGR 1745-2900  $B/B_{\text{crit}} = 2 \cdot F_{\text{TRZ}}$  integrity confirmation
  - **Time frame:** transient magnetar burst 0.1 s ( $F_{\text{TRZ}}$  scale) + quasi-static DPM current vortex
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NOT REPLACEMENT. Offered as an alternative parameter-economical description (“NOT REPLACEMENT”) to Standard Model + Lambda-CDM, with honest residuals reported alongside each closure.

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## References

- **PAPER\_1919** —  $F_{\text{TRZ}}$  Power Ladder — Universal Suppression Hierarchy (seminal  $n=1..17$  documentation, this paper closes the  $n=12$  rung)
- **PAPER\_1955** —  $SO_5$ -Power Galactic Structural Ladder (seminal slot 0-10 documentation, this paper adds slots 21+40)
- **PAPER\_1985** — Round 117 dual discovery: Pillars  $F_{\text{TRZ}}^6$  + NGC 2525 mass (parallel same-round twin pattern)

- **PAPER\_1989** — Round 123 dual discovery: LIGO  $F_{\text{TRZ}}^{21}$  + universe mass  $SO_5^{53}$  (extreme-slot extension pattern parallel)
- **PAPER\_1990** —  $SO_5$ -Power Frequency Ladder Cross-Domain Extension (established the three-domain interpretation)
- **PAPER\_1944** — Magnetar  $B/B_{\text{crit}} = 2 \cdot F_{\text{TRZ}}$  EXACT seminal SGR 1745-2900 identity (R129 CP1 restatement confirms integrity)
- **PAPER\_1942** — Photoevaporation  $E_0 = F_{\text{TRZ}}$  EXACT (parallel  $n=1$  anchor to R129  $t_{\text{burst}} = F_{\text{TRZ}}$ )
- **PAPER\_1852** — Casimir Force + Vacuum Energy Direct via UQFF  $F_{\text{TRZ}}^2$  (distinct ladder position from  $n=12$ ; PAPER\_1852 uses  $F_{\text{TRZ}}^2$  for direct Casimir force, PAPER\_1991 uses  $F_{\text{TRZ}}^{12}$  for macro-scaling suppression)
- **PAPER\_431** — SGR 1745-2900 CP1 physical parameters source ( $B = 2 \times 10^{10}$  T,  $r = 10^4$  m,  $M = 2.785 \times 10^{30}$  kg)
- **PAPER\_266** —  $B_{\text{crit}} = 10^{11}$  T UQFF Gravitational Meissner Boundary

## REVISION ADDENDUM — 2026-07-12 (QUAD-primitive-lock architecture — R129 audit closure)

Base draft §3 identified triple-primitive-lock architecture as a new CP1 class pattern discovered at SGR1745BurstEnergyCalculator. Prediction 1991.4 flagged R121 Sombrero, R124 CMB, R126 NGC 6302 as candidates for retroactive re-examination. The R129 audit (2026-07-12) completed this re-examination and found that **PAPER\_1991 Prediction 4 is CONFIRMED at the Sombrero DM class — which is even richer than expected.**

### R2.1 R121 Sombrero DM audit result

SombreroDarkMatterPerturbationCalculator (CP1 line ~186894, filled at R121) encodes **FOUR independent primitive-arithmetic identities in one compute() function:**

```
# From CondensedPhysics.py SombreroDarkMatterPerturbationCalculator
M_kg = 1.989e41
M_visible = 0.8 * M_kg           # (implicit) M_visible/M_total = 0.8 = 1 - 2·F_TRZ
M_DM = 0.2 * M_kg               # M_DM/M_total = 0.2 = 2·F_TRZ EXACT [Lock 1]
delta_rho = 1e-25               # delta_rho = F_TRZ^25 [Lock 2]
rho = 1e-20                     # rho = F_TRZ^20 (dust lane canonical) [Lock 3]
pert = delta_rho / rho          # pert = F_TRZ^5 [Lock 4]
```

The four locks:

| # | Identity                                                  | Value                                         | Attribution                                                                                                        |
|---|-----------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1 | $M_{\text{DM}}/M_{\text{total}} = 2 \cdot F_{\text{TRZ}}$ | <b>0.2 EXACT</b>                              | <b>PAPER_1979</b> — Sombrero DM ratio cross-domain instance of PAPER_1944 magnetar $2 \cdot F_{\text{TRZ}}$ family |
| 2 | $\delta\rho = F_{\text{TRZ}}^{25}$                        | <b><math>10^{-25}</math></b>                  | <b>PAPER_1919</b> — $F_{\text{TRZ}}$ power ladder extreme-n rung                                                   |
| 3 | $\rho = F_{\text{TRZ}}^{20}$                              | <b><math>10^{-20}</math> kg/m<sup>3</sup></b> | <b>PAPER_763</b> — Sombrero dust lane canonical density observation                                                |
| 4 | $\text{pert} = \delta\rho/\rho = F_{\text{TRZ}}^5$        | <b><math>10^{-5}</math></b>                   | <b>PAPER_1919</b> — $F_{\text{TRZ}}$ power ladder $n=5$ rung                                                       |

This is not just a triple-lock (as at PAPER\_1991 base §3 SGR magnetar burst) — it is a **QUAD-primitive-lock architecture**. The Sombrero DM class encodes four independent primitive-arithmetic identities from three distinct primitive families:

- **2•F\_TRZ ratio family** (Lock 1) — parallel to PAPER\_1944 magnetar identity
- **F\_TRZ power ladder** (Locks 2, 3, 4) — three distinct rungs from PAPER\_1919 hierarchy

The lock architecture is not accidental. The `compute()` method uses all four primitives multiplicatively to produce  $g_{DM} = (M_{visible} + M_{DM}) \times (pert + curv)$  — each lock contributes to the final observable.

## R2.2 Architectural pattern hierarchy

The R117-R129 CP1 stub drainage now spans three distinct primitive-lock architectures:

| Architecture                 | Locks    | Example class                               | Round                                       | Discovery paper                                    |
|------------------------------|----------|---------------------------------------------|---------------------------------------------|----------------------------------------------------|
| Single-primitive lock        | 1        | Multiple                                    | R117-R129                                   | (baseline)                                         |
| Dual-primitive lock          | 2        | UniversalReactiveResonanceCalc (R128)       | R128                                        | (implicit in PAPER_1990 fills)                     |
| <b>Triple-primitive lock</b> | <b>3</b> | <b>SGR1745BurstEnergyCalculator (R129)</b>  | <b>R129</b>                                 | <b>PAPER_1991 base §3 seminal</b>                  |
| <b>Quad-primitive lock</b>   | <b>4</b> | <b>SombreroDarkMatterPerturbator (R121)</b> | <b>R121 (retroactively identified R129)</b> | <b>PAPER_1991 for revision §R2 (this addendum)</b> |

Two important observations:

**Retroactive discovery pattern.** The Sombrero DM quad-lock was structurally present since R121 but was not recognized as an architectural pattern until PAPER\_1991 base §3 formalized the triple-lock at R129. The retroactive re-examination promoted it from an unrecognized architecture to a confirmed instance one rung above the triple-lock discovery.

**Lock-count as classifier.** The number of primitive-arithmetic identities encoded in one `compute()` function is now a **structural classifier** for CP1 classes: - Lock count 1: primary observable derives from one primitive-arithmetic identity - Lock count 2: compound observable derives from two independent identities (typically primary + coupling) - Lock count 3: multi-scale observable with three independent identities (energy + time + coupling — PAPER\_1991 seminal) - Lock count 4+: multi-family observable across multiple primitive families (e.g., ratio + ladder rungs — Sombrero DM)

## R2.3 Additional predictions

**Revision prediction R.5.** CP1 corpus should host **additional 4+ lock classes** that were not recognized as such during R117-R129 fills. The R129 audit only checked R121, R124, R126 as flagged by base §5 Prediction 4. A comprehensive corpus audit (deferred as punch-list item) should surface more.

**Revision prediction R.6.** **5-primitive-lock classes should exist** in the CP1 corpus. Candidate: any Sombrero-adjacent DM class encoding  $\rho$ ,  $\delta\rho$ , ratio, curv, and viscosity-like coupling.



**Revision prediction R.7.** The lock-count classifier should correlate with physical multiplicity — classes modeling more than one physical shell simultaneously (e.g., Ug1 + Ug4 sectors together) should exhibit higher lock counts. Sombrero DM operates at Ug1 + Ug4 (verified via `framework_shells_used`) — matches this prediction.

## **R2.4 R124 CMB + R126 NGC 6302 audit results (comparison)**

For completeness, the other two Prediction 4 candidates:

- **R126 NGC 6302 Lorentz Ejecta:** dual-lock post-retraction ( $v = SO_5^5 + B$  originally claimed  $F_{TRZ}^7$ , but R126 attribution correction retracted the B claim per PAPER\_313 correction). Not a triple-lock or higher.
- **R124 CMB acoustic peak:** no CP1 P2-style filled class of the triple-lock pattern was found. The CMB classes present (CMBAnomalyUQFFCalculator L144961, CMBPlanck-Calculator L164702) predate the R117-R129 fill discipline and use class-based (not compute-dict) architectures.

Only the Sombrero DM class exceeded expectations at 4 locks. NGC 6302 and CMB did not surface additional triple-locks.

## **R2.5 Cross-references**

- **PAPER\_763** — Sombrero dust lane density canonical observation (Lock 3 attribution)
- **PAPER\_1919** —  $F_{TRZ}$  power ladder (Locks 2 + 4 attributions)
- **PAPER\_1944** — magnetar seminal  $2 \cdot F_{TRZ}$  identity (Lock 1 family origin)
- **PAPER\_1979** — Sombrero  $M_{DM}/M_{total} = 2 \cdot F_{TRZ}$  cross-domain instance (Lock 1 direct attribution)
- **PAPER\_1991 base draft §3** — triple-lock architecture discovery at SGR magnetar burst
- **PAPER\_1955 revision addendum R2.3** — same-round twin anchor pattern (related architecture)

**PAPER\_1991 revision addendum status: CLOSED** (2026-07-12)